

From hypothetical selection machinery to a pure prediction system

What the system was, what its own record proved, what replaced it on July 8, and why the replacement is structurally better. Removed modules remain recoverable in git history (commits `1909945` → `d5ae2b7`).

WAS

A selection loop on top of the projections

Through July 7 the pipeline's product was hypothetical multi-pick *sets*: a combination builder priced 2–5 pick sets with platform multipliers, computed an expected-value figure, and sized notional positions against a \$1,000 balance. Suppression thresholds meant a row only surfaced if its stated probability cleared the multiplier-implied bar plus a margin — days with no qualifying combination produced *no output at all*. Results were tracked in dollars, and the dashboard announced states: LIVE, WAITING, "picks locked".

Critically, the raw strikeout projection (`expected_ks`) was trusted at face value — whatever confidence the distribution produced was displayed and acted on.

What the record proved

DATE	ROWS	STATED	REALIZED	GAP
Jul 5	pitcher strikeouts	69.0%	55.6%	-13.5 pts
Jul 7	pitcher strikeouts	71.7%	46.7%	-25.0 pts

One of eight hypothetical sets landed; the notional record ran −\$70 (−53.8%), and the drift widened day over day. Each response — dispersion re-fit, probability ceiling — treated a symptom.

$$s = 0.00$$

The root cause landed July 8. Tested on **164 frozen pre-game projections** (logged before their games, settled against official boxscores — leakage impossible), the model's chosen sides realized **50.4%** while stating 61.3%; the rows stated at $\geq 65\%$ realized 55.9%. Fitting $\mu' = \text{line} + s \cdot (\mu - \text{line})$ gave $s = 0.00$: the projection's disagreement with the published line carried no walk-forward information. The line alone predicted better than the line plus the model.

The machinery was sizing hypothetical positions on confidence the data never supported — and its suppression thresholds guaranteed that whenever it did speak, it spoke from the model's most over-confident tail.

NOW Numbers in, accuracy out

WAS

- Output only when thresholds cleared; silent days
- Raw projection trusted at face value
- \$-denominated record, multiplier math
- LIVE / WAITING state banners
- RotoWire disagreement removed rows
- New sources usable immediately

NOW

- Every matched row scored, always — ranked, never suppressed
- Probability from a line-anchored mean; coefficient fitted on frozen data
- Hit rate + stated-vs-realized calibration, graded from official boxscores
- Factual freshness line, no states
- RotoWire shown as annotation beside the numbers
- New sources earn a coefficient on frozen snapshots first

The point prediction and the probability are deliberately decoupled: the displayed prediction is the raw model output (a real number, e.g. 6.83 K), while the probability reflects only what settled frozen data has supported. With s currently 0.00, strikeout probabilities honestly sit near 50% — a statement of no information, printed rather than hidden.

Why this is better

- 1 **The probabilities are honest.** Stated confidence is bounded by frozen out-of-sample evidence, and the public calibration line shows the gap daily — a 73%-stated / 56%-realized failure can no longer happen silently.
- 2 **Output can't be selection-biased.** The old thresholds meant the system only spoke when most over-confident. Scoring the whole board makes the graded sample the whole board.
- 3 **The measurement loop is leakage-proof by construction.** Every fit input is frozen before its game. Both decisive findings — the $r \rightarrow 500$ leakage incident and $s = 0.00$ — came from enforcing exactly this rule.
- 4 **New sources earn influence numerically.** FantasyPros consensus (frozen daily since Jul 8) feeds nothing until it fits a coefficient above zero walk-forward — the audit the old source failed.
- 5 **The scoreboard matches the product.** A projection system is good when stated 65% realizes 65%. Hit rate and calibration measure that directly; dollar figures measured it through set-construction choices unrelated to model quality.
- 6 **The vocabulary is enforced, not aspirational.** A pre-commit hook blocks the removed terminology from ever being committed again.

What carried over unchanged

The parts that were already measured stayed: the Negative-Binomial strikeout distribution with frozen-fitted dispersion ($r = 16.6$, PIT-validated), the batter baseline with matchup and platoon adjustment under its 70% ceiling, the day-factor correlation model ($\tau \approx 0.08$), board capture with the stale-board guard, the hourly cron, and the frozen daily snapshots.

What better looks like in the record

stated $\approx$ realized

Short term, hit rate near 50–55% with small gaps is **correct behavior** for a source that currently adds nothing beyond the line. Within one to two weeks, enough frozen consensus days settle to fit the FantasyPros coefficient; if it fits above zero, strikeout confidence rises by a measured amount. The public calibration line is the contract: when stated and realized diverge, a coefficient gets re-fitted on frozen data. Numbers move — never modes.

Full technical version: [docs/TRANSITION.md](#) in the repository · dashboard: [fantasy.perfecthold.online](#) · prepared 2026-07-08